Program overview

14-Dec-2022 11:00

Year	**2022/2023**
Organization	**Aerospace Engineering**
Education	**Minors Aerospace Engineering**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors LR 2022							
LR-MI-176-22 Offshore Wind Energy							
AE3512-20	Asset Management	5					
AE3513	Integratieopdracht	6					
AE3515	Basics of Aeroacoustics for Wind Energy	3					
AE3516A	Fundamentals of Wind Energy I	3					
AE3516B	Fundamentals of Wind Energy II	3					
CT3101	Project Management Basics	5					
TBM024B	Introduction to Energy Systems	5					
LR-MI-228-212 Airport Development	**LR-MI-228-22 Airport Development**						
AE3501-19	Air Transportation	3					
AE3502-14	Airport Planning, Design and Operations	4					
AE3503	Strategic Planning for Airport Systems	6					
CT3080LR	Landside accessibility of Airports	6					
IO3818	Designing an Airport Ecosystem	6					
TB241TB	Logistics 2	5					
LR-MI-232-22 Space Missions							
Standard programme							
AE3522UL	Fundamentals of Astronomy	4					
AE3523	Planetary Sciences	3					
AE3524	Spacecraft Design and Astrodynamics	5					
AE3526	Project Design and Analysis	2					
AE3527UL	Scientific Case Study	4					
AE3528	Engineering Case Study	4					
AE3529	Space Instrumentation	5					
CT3533	Earth Observation	3					
Programme for Astronomy students (Leiden University)							
AE3523	Planetary Sciences	3					
AE3524	Spacecraft Design and Astrodynamics	5					
AE3524E	Spacecraft Design and Astrodynamics Extended	4					
AE3526	Project Design and Analysis	2					
AE3527UL	Scientific Case Study	4					
AE3528	Engineering Case Study	4					
AE3529	Space Instrumentation	5					
CT3533	Earth Observation	3					

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LR-MI-176-22 Offshore Wind Energy	
Minor Coordinator	Dr.ir. A.C. Viré

AE3512-20	Asset Management	5
Responsible Instructor	Dr. D. Zappalá	
Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr. R.M. Groves	
Instructor	Prof.dr. S.J. Watson	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic coding knowledge (Python/Matlab or similar programming languages).	
Course Contents	Asset management concerns all activities related to the primary company values relative to asset ownership and use. It is most visible in execution of the strategic plan which an organization must have in place to optimally manage its assets for realizing its core values. Optimal asset management incorporates addressing the balance between performance, risk and costs over the entire asset lifecycle. For many asset-intensive companies, asset management is the primary business process. In recent years, various standards have been developed for asset management which are now being implemented in industry. An important element in these standards is that the optimal balance between costs, performance and risks are addressed at the strategic level (from a systems perspective) and covers the entire portfolio of assets. To reach the aforementioned optimal balance, asset-intensive companies require significant volumes and variety of data (e.g. asset condition data, contractual data, operational data, etc.). This course provides basic systems-oriented asset management concepts and tools, and introduces students to the principles, challenges and solutions faced by the wind industry. Effective asset management is crucial for optimizing the lifecycle cost of offshore wind energy which is a rapidly developing industry sector. The course combines state-of-the art theory with concrete examples and practical applications to wind energy assets. During this course students have the unique opportunity to directly hear from and engage with world-leading industry and academic experts on wind farm asset management. They are introduced to the current practice and innovation in industry as well as to the state-of-the-art and latest developments of the research in the field. The study material of the course includes a selection of research articles, which is regularly updated as novel research is generated. Hands-on assignments give students the opportunity to apply what they learn during the lectures to real wind farm management-related problems.	
Study Goals	In this course, you will learn the foundation of the theory of modern systems-oriented asset management and how to extend and apply this knowledge to current and future offshore wind farms in order to optimise their life-cycle costs. By the end of this course, you should be able to: L01: Explain the primary asset management concepts, tools and policies and their applications to offshore wind farms. LO2: Describe the key concepts (RAMS) governing the management of wind assets. LO3: Compare maintenance and logistics strategies and policies (corrective, preventive, predictive, condition-based maintenance) and evaluate the most cost-effective and intelligent approach for offshore wind farms. LO4: Critically assess state-of-the art monitoring data acquisition (SCADA and CMS systems) and analysis tools of wind assets and discuss future improvements in these areas. LO5: Formulate strategies to mitigate the main issues (from design to end-of life) affecting the life-cycle costs of offshore wind turbine blades.	
Education Method	The course consists of class lectures. For a number of topics, guest lecturers from academia and industry are invited.	
Assessment	The final assessment consists of a written exam (open questions). Additionally, a maximum 0.5 bonus point (out of 10 points) can be earned by performing the hands-on assignments during the course. The bonus point is only valid for the regular exam of that academic year, not for the resit.	

AE3513	Integratieopdracht	6
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The assignment consists in answering key questions in the future development of offshore wind energy. The questions can relate to technical challenges, as well as environmental and management aspects. As far as possible, the assignment is done in collaboration with a company or a governmental body.	
Study Goals	At the end of the assignment, the student should be able to: - integrate knowledge from the different courses and sources, - reflect on challenges and trends for the future of offshore wind power, - report the work in a coherent written document and oral presentation of high academic standards, - work in a team to answer questions of industrial interest.	
Education Method	Independent teamwork.	
Assessment	Written report and online oral presentation.	

AE3515	Basics of Aeroacoustics for Wind Energy	3
Responsible Instructor	Dr. D. Ragni	
Responsible Instructor	Dr. F. Avallone	
Responsible for assignments	Dr. D. Ragni	
Co-responsible for assignments	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching the world of sustainable energies. In wind-energy, new wind farms are designed with extremely large rotors. Off-shore wind-farms are additionally built relatively closer to the coast line and to populated areas, in order to benefit from those locations where the winds are relatively stronger. The aim of this course is to provide the students with a fundamental knowledge about noise sources with full focus and attention on the problem of wind-turbine noise. The course is structured in three main areas: 1) Introduction to noise sources and sound: This module introduces the basic knowledge about sound, sound pressure levels, sound power level and wave propagation. 2) Airfoil self-noise: In this module the students will study the fundamental sources of noise in airfoils operating at high-Reynolds number, with extension to blade noise for wind-turbine applications. 3) Wind-turbine noise The third area of the course deals with the theory behind wind-turbine noise. Experimental and numerical applications will be taken into account with an industrial view to the problem and to the most interesting noise-mitigation strategies.	
Study Goals	The course is meant to give the students a basic knowledge of wind-turbine noise for on/off shore applications. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Know what sound is and how noise is generated - Describe the most important sources of noise for airfoils and how to translate that to a wind-turbine blade - Analyze the most important sources of noise for wind-turbine rotors and their consequences for on/off shore applications	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Wind-Turbine Noise, Authors: Wagner, Siegfried, Bareiß, Rainer, Guidati, Gianfranco - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of a written exam with multiple choice and open questions.	

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

AE3516B	Fundamentals of Wind Energy II	3
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy production from the perspective of a wind turbine Original Equipment Manufacturer (OEM). The main questions covered in the course are: - Why harvest offshore wind energy? - What are key components of a wind turbine? - What are objectives and steps in the wind turbine design process? - What are key technical, economic and social acceptance related challenges and opportunities for offshore wind energy? - What are current trends and emerging technologies in offshore wind energy?	
Study Goals	At the end of this course, in the context of offshore wind, the students will be able to: - Identify different stakeholders and reflect on differences in their interests. - Explain the basic components of a wind turbine. - Explain the objectives of the wind turbine design and name the steps of the process. - Identify technical, financial and social acceptance related challenges and opportunities. - Reflect on current trends and emerging technologies.	
Education Method	The teaching method uses a standard classroom approach with activation elements. This means that, in class, exercises and feedback are given to or collected from the students. Supplemental material for further study is provided after each class.	
Assessment	The final assessment is a written exam.	

CT3101	Project Management Basics	5
Responsible Instructor	M.G.C. Bosch-Rekveldt	
Instructor	Prof.dr.ir. M.J.C.M. Hertogh	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is an important cornerstone of the minor Project Management: from Nano to Mega. In this course the general PM principles will be taught such as: iron triangle, trade-offs, scoping, scheduling, risk management, monitoring and control, project start-up, project close-out. Also the more strategic context of a project in an organization is discussed including portfolio management and the business case of a project. Throughout, the project life-cycle acts as a basis: A. Conception B. Definition C. Execution, contracting, implementation D. Handover Trade-offs between time, cost, quality, and scope are extensively discussed, after discussing the different axes in detail: -SCOPE axis - setting the project boundaries, WBS; -TIME axis - planning & scheduling, network scheduling, critical path method (CPM), Program Evaluation and Review Technique (PERT), resource constraints; -COST axis - estimating, project cost accounting, scheduling and forecasting for costs; -QUALITY axis - risk, TQM, six sigma, other techniques. Attention is paid to both front-end development and project execution. During the project execution the execution team utilizes all the schedules, procedures and templates that were prepared and anticipated during prior phases. Unanticipated events and situations will inevitably be encountered, and the Project Manager and Project Team will have to deal with it (change management). Overall: People are key in successful project management!	
Study Goals	The course is aimed at getting to know the basics of project management and being able to apply the obtained knowledge on real life cases. More specifically, after following this course the student is able to 1. Explain the difference between projects, programs and portfolios. 2. Explain how value is developed in projects throughout the project life cycle (PLC). 3. Select and reflect upon an appropriate organisation structure for a project. 4. Construct a project planning using MS Project /PertMaster. 5. Create a risk register based upon a qualitative risk analysis. 6. Explain the role of change management techniques in project management.	
Education Method	This course will be mainly based on lectures (/guest lectures) with smaller assessed weekly assignments for elaborating on the various topics. We will also use study materials from the MOOC Management of Engineering Projects: Preparing for success!	
Course Relations	This course is part of the minor Project Management: from Nano to Mega (CT-MI-174). Also students of the minor Offshore Wind Energy follow this course. Non-minor students might participate as well, but always should contact the lecturer upfront.	
Assessment	The assessment consists of - Written exam (half knowledge questions, half questions related to a casus) - Assignments (obligatory), further specified during lecture 1, which all should be completed before the final grade of the exam can be obtained. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently).	
Tags	Business Project planning / management	
Expected prior Knowledge	No specific prior knowledge required.	
Academic Skills	This course will develop your writing and reflecting skills. And of course your project management skills!	
Literature & Study Materials	Management of Engineering Projects - People Are key. Hans L.M.Bakker and Jaap P. de Kleijn, editors. NAP - The Process Industry Competence Network. ISBN/EAN: 978-90-812162-0-3 This book can be ordered (student-price) via VSSD.	
Judgement	The weekly assignments will be graded with fail - sufficient - good. All weekly assignments need to be at least sufficient: for a "fail", rework has to be done. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No materials are permitted during the Exam, except Dutch-English (or vice versa) dictionaries. Also a simple calculator can be used.	
Collegerama	Yes	

TBM024B	Introduction to Energy Systems	5
Responsible Instructor	Dr.ir. M.D.M. Pérez-Fortes	
Module Manager	Prof.dr.ir. C.A. Ramirez Ramirez	
Instructor	T.J. Wiltink	
Instructor	J.T. Manalal	
Responsible for assignments	Dr.ir. M.D.M. Pérez-Fortes	
Co-responsible for assignments	Prof.dr.ir. C.A. Ramirez Ramirez	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This is a core course to the Offshore Wind Energy Minor from AE.	
Expected prior knowledge	Basic math, physics, and chemistry/technology knowledge are required. Perform basic mass and energy balances.	
Course Contents	By the end of this course, you should be able to: Discuss the characteristics of an energy system at different levels of aggregation. Calculate technical, environmental, and economic indicators at different levels of aggregation. Estimate system performance based on calculated indicators and contextual information. Compare energy policies and market issues that would impact a selected system at a specific level of aggregation.	
Course Contents Continuation	Energy systems embrace multiple technologies, sectors, and perspectives. In this course, you will understand the multiscale (i.e. different levels of aggregation, i.e. equipment, household/industry, energy chain, sector, region, country, international, world) and multi-perspective (i.e. technical, environmental, economic, and governance) nature of the analysis of energy systems and you will be able to (i) apply methodologies to estimate technical, environmental and economic performance and (ii) recommend options to improve performance at different levels of aggregation. The course provides the foundation for the understanding of the main elements that characterise energy systems. Special attention is given to global developments, and past, current, and future trends. This includes the current transition from a fossil-fuel based system to a climate-neutral energy system and related potentials and scenarios. Energy systems analysis is a field where good argumentation skills are essential, so in addition to calculations, there is a significant focus on critical thinking, discussion, and reflection.	
Study Goals	By the end of this course, you should be able to: Discuss the characteristics of an energy system at different levels of aggregation. Calculate technical, environmental, and economic indicators at different levels of aggregation. Estimate system performance based on calculated indicators and contextual information. Compare energy policies and market issues that would impact a selected system at a specific level of aggregation.	
Education Method	Lectures and self-study/homework. The course counts with guest lectures from experts working in the different aspects of energy systems. The book Introduction to Energy Analysis by Kornelis Blok and Evert Nieuwlaar, third edition, is used to complement the content of the lectures and for homework problems. Each week there are two interactive lecture classes (2 hours each) and a problem-solving session (1 hour). There will be room for Q&A. To take the most of this course you should participate in class and do the homework.	
Literature and Study Materials	Introduction to Energy Analysis by Kornelis Blok and Evert Nieuwlaar, third edition. Lecture slides (Brightspace).	
Assessment	The assessment consists of an individual assignment and a written open-book exam. The individual assignment has a weight of 40%. The exam takes place at the end of the course (week 10, and lasts for 3 hours) and has a weight of 60%. To pass the course, individual assignment and exam must be graded with at least a 5.0 (minimum combined grade of 5.75). In case of unforeseen circumstances or measures resulting from COVID-19, the written open-book exam on campus will be replaced by a timed take-home open-book exam.	
Permitted Materials during Tests	Course book, a 2-page (4 sides) formula sheet and a calculator.	
Remarks	English is the language used in the course and in the exam.	
Targetgroup	Students from the Offshore Wind Energy Minor, to get an overview and understanding of how energy systems, i.e. the whole energy-based supply chain, are designed, and how these can and have to be transformed towards a sustainable future. The course provides basic knowledge for strategic and tactical planning. It is an overarching course that goes into energy consumption, electricity generation, sustainability, and climate change mitigation.	
Category	BSc level	

Year	**2022/2023**
Organization	**Aerospace Engineering**
Education	**Minors Aerospace Engineering**

LR-MI-228-212 Airport Development	
Minor Coordinator	Ir. P.C. Roling

AE3501-19	Air Transportation	3
Responsible Instructor	Ir. P.C. Roling	
Instructor	Dr.ir. B.F. Lopes dos Santos	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course offers an introduction to the various aspects of air transportation, including air law, regulatory frameworks, airline economics, airline network and fleet planning, air transport safety, airport development, environmental impact, and air traffic management.	
Study Goals	Identify the characteristics and trends in the airline industry Explain the role of air transport in the global economy Identify and describe the economic and technical regulations and standards pertaining to organisations within the air transport industry. Identify and describe the structure of airline costs Describe the economic concepts of supply, demand, pricing and market structures Explain the key characteristics of different airline business models including the role of the airport. Explain the different network and fleet strategies adopted by airlines Explain key demand and supply factors affecting the air cargo sector	
Education Method	Lectures with possibly some exercises.	
Literature and Study Materials	R. Doganis, Flying off course, the economics of international airlines, London, Routledge, 2010, 4th Ed.	
Assessment	Digital exam	

AE3502-14	Airport Planning, Design and Operations	4
Responsible Instructor	Ir. P.C. Roling	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Airport Planning, Design and Operation aims to provide students with an understanding of the complex interrelationships and interactions among airport capacity, airport demand, policy changes, investments, and environmental issues and the effects that changes in any of these can have on airport profits and performance	
Study Goals	Being able to Estimate the capacity of any airport configuration and understand the influence of weather, aircraft mix, and other operational parameters on capacity Determine runway requirements for a given aircraft type Estimate the (future) demand for transport of both passengers and freight Estimate the flight delays at an airport given the capacity of the facility and daily demand factors Analyze the noise impacts of aircraft in the vicinity of airports using computer models Estimate the geometric design characteristics of an airport including taxiways, aprons and runways Estimate siting criteria for new airports including terminals Conduct a Cost-Benefit analysis for new airport infrastructure developments Present and defend the policy recommendations consistent with the long-terms vision for the airport	
Education Method	Lectures	
Reader	Richard de Neufville and Amedeo R. Odoni, Airport Systems: Planning, Design, and Management, McGraw-Hill, 2003	
Assessment	Computer-based exam (80% of grade) and homework assignments (20% of grade)	

AE3503	Strategic Planning for Airport Systems	6
Responsible Instructor	Ir. P.C. Roling	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The airport business is high-paced, dynamic and unpredictable. Airport decision makers are faced with the continual challenge of having to make strategic plans and business decisions regarding infrastructure, market positioning, and commercial activities at the airport. The final project will explore the various aspects of strategic planning for airport systems	
Study Goals	The major goal of the capstone project is that it aims to provide students with an understanding of the complex interrelationships and interactions among airport capacity, airport demand, policy changes, investments, security and environmental issues and the effects that changes in any of these can have on airport profits and performance. Using a policy analysis/system-modeling framework, the full range of airport strategic planning issues will be addressed.	
Education Method	The final project involves a lab exercise in which each team of (5) students is asked to design and operate your own airport in a simulated environment based on the in-house developed Airport Business Suite (ABS). ABS is an integrated computational platform that will support policy and business decisions relating to airport development, through an integrated impact analysis. The lab exercise is directly linked to the lectures of the course Airport Planning, Design and Operation. Next to this, other tool s may also be part of the project.	
Course Relations	AE3502 must have been followed before taking part.	
Literature and Study Materials	Richard de Neufville and Amedeo R. Odoni, Airport Systems: Planning, Design, and Management, McGraw-Hill, 2003	
Assessment	The final project is to be concluded with a written report, along with a presentation in a mini-symposium. There is no written exam. The project is graded based on the quality of the report. A peer evaluation will be used within each group.	

CT3080LR	Landside accessibility of Airports	6
Responsible Instructor	Dr.ir. A.M. Salomons	
Responsible Instructor	Dr.ir. A. Gavriilidou	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course offers an introduction to all aspects of landside accessibility through assessing performances of transport modes and operational systems, related to transport to and from the airport. This includes elements of infrastructure and transport service networks, for which have to be considered: demand, capacity, spatial layout, economics, environmental and social impacts, local infrastructural development, and the complex integration of airports into regional and national transport system(s). Air transport system - components and operations Airport and airport system - components and operations, including crowd monitoring and management in airports Landside access modes and systems as a component of an airport: conventional and advanced road & rail-based - components, operations, performances Examination - analysis, modelling, and estimation - of performances of airport access modes and systems - infrastructural, technical/technological, operational, economic, environmental, social, policy Control and management of landside access modes and their systems: introduction into ITS (Intelligent Transport Solutions) Planning and design of landside access modes and systems; Developing an airport as the multimodal transport node; Airport City, agglomeration effects of airports, and urban planning and development around airports.	
Study Goals	After successful completion of the course, the student is able to: - identify the characteristics - components and performances - of transport modes and their systems providing landside access to an airport - analyze, model, evaluate, plan, and design of airport landside access modes and their systems means by multidimenisonal examination of their performances and related influencing factors; - analyze and discuss the role of airports as multimodal transport modes in transport networks - analyze and discuss urban planning and developments around airports as well as their agglomeration effects	
Education Method	Lectures (on campus if Covid allows) and assignment	
Assessment	1) Written exam (65%) 2) Assignment (to be handed in before the exam) (35%)	
Tags	Building & Spatial Development Modelling Policy Analysis Project planning / management Research Methods Transport & Logistics	
Expected prior Knowledge	No	
Academic Skills	Students are expected to exercise the following academic skills: Interpretation, writing review, judgemental skills, reasoning / arguing, writing concise and clear reports.	
Literature & Study Materials	BrightSpace postings of lecture notes and presentations; book "Landside Accessibility of Airports" by Milan Janic	
Judgement	1) Written exam (65%) 2) Assignment (to be handed in before the exam) (35%) (5% - Midterm presentation; 5% - Group peer review; 25% - Assignment report)	
Permitted Materials during Exam	To be announced	
Collegerama	No	

IO3818	Designing an Airport Ecosystem	6
Course Coordinator	Ir. M.F. Beets	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The course Designing in Airport Ecosystem is built around an airport system and mobility. It is a research and design course with Schiphol as the main stakeholder. Other stakeholders join as well. You work on a real case for a client in one of busiest, main airports of Europe with around 70 million passengers a year. You study the airport, passengers from home to destination and back. You will explore about touchpoints with the different stakeholders and their interests in passengers. In doing research with users you will find hick-ups in speed or experience, that can be improved. You may create a seamless passenger journey using current technology that meets the interests of the user and the client and relevant other stakeholders. Depending on the case you will design product/services, like: digital service, baggage-, tax free- or catering points. You work in a group of 4 students of different faculties in a multi-disciplinary team.	
Study Goals	You will get insight in the challenges of what is showing up while designing operations at an airport. Learning about the different stakeholders involved and their needs/interests. You will get insight in designing from a passenger perspective. You will meet and learn about airport operations. Learning goals : LO1: analyze the context and the role of stakeholders within an airport ecosystem and create a frame work for design. LO2: to integrate desirable, feasible and viable aspects in a concept proposal and prototype. LO3: to apply the double diamond innovation process and tools and methods to design. LO4: to present relevant issues and design decisions of your work in a in a argued and concise way to stakeholders and others. LO5: to reflect on the results in relation to the case owner and the design goal and criteria.	
Education Method	The course is structured into 2 parts. Part1- Is working on your project brief and explore research and observe, to find challenges and user needs/ problems in the context. You will end up with a design brief which includes a (re) defined problem statement. Part 2 is a developing phase, generating ideas and concepts. You finally end up with a concept and prototype of a service/product what meets the requirements of the client, the user and other stakeholders. You will finally present your work for the client, peers and coaches. Working style: -Guest lectures and lectures to inform you about the airport system and design methods. -Work on your project with your group. -Coach meetings with feedback once a week, clients are invited several times during the course. -Presentations for the client and peers. -Exhibition/presentations in the last week, end of all projects.	
Literature and Study Materials	Relevant books and reading materials will be presented troughout the course. Links will be available on Brightspace.	
Assessment	Your final individual grade will assessed by using a rubric. You will deliver: -frame report (research findings and designbrief) -design report (itteration and design results) -presentation and prototype -reflection	

TB241TB	Logistics 2	5

Module Manager	Dr. S. Fazi
Co-responsible for assignments	Dr. J.A. Annema
Contact Hours / Week x/x/x/x	4/0/0/0 excl 10 hours of self study
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Course Contents	According to the Council of Supply Chain Management Professionals logistics is defined as the process of planning, implementing, and controlling procedures for the efficient and effective transportation and storage of goods including services, and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements. Logistics activities should be in line with other business activities such as marketing and purchasing, in order to create sustainable competitive advantages for individual companies and service providers, and for the supply chain as a whole. Given the strategic importance of logistics for individual organizations and supply chains, it is vital to know how the performance of logistics activities is measured, and how these activities can be modelled and improved. Therefore in this course different logistics aspects are discussed and analyzed. A selection of quantitative methodologies are introduced to model and provide solutions to a wide range of relevant problems in logistics. We focus on two classes of logistics problems: Logistics problems within organizations: forecasting and demand modeling; inventory management; clustering and classification of items. Distribution problems. Methodologies include analytical models, linear programming models, basic data mining techniques.
Study Goals	At the end of this course you will be able to: Analyze relevant aspects of logistics and transportation system: inventory, routing, location, data; Formulate (and solve) a selection of relevant logistics problems; Select and apply an appropriate methodology for a given logistics problem; Find possibilities to improve/adapt a logistics system through data and analytical models; Model operational aspects through mathematical linear programming models; Understand the integrative role of logistics function in and between organizations;
Education Method	Lectures Assignments Self-study It is expected that for academic year 2022/2023 the educational activities will be held on-campus. Consultation hours will be held both online and on-campus.
Literature and Study Materials	A selection of chapters from: 1) Reid, R. D., Sanders, N. R., Operations Management: An Integrated Approach, 5th edition, Wiley, 2013. 2) Daganzo, C.F., Logistics Systems Analysis, 4th edition, Springer-Verlag, 2010. 3) Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. Data mining and analysis: fundamental concepts and algorithms. Cambridge University Press, 2014. 4) A booklet provided at the start of the course 5) Slides and exercises
Prerequisites	Logistiek 1, Basic mathematics
Assessment	The course consists of two parts with the following weighting: Logistics problems within organizations: 50%. Distribution problems: 50%. Grading: One written exam: 80% Assignments: 20% The passing grade is 5.8, calculated as: 0.8*(final exam) + 0.2*((Assignment 1 + Assignment 2)/2). Minimum grade for final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 -> (Assignment 1 + Assignment 2)/2 >= 5.0 In case of unforeseen circumstances or measures resulting from COVID-19, the final exam will take place in an online format: a take-home exam with a limited duration. The student will receive the exam via email at the start of the exam session and must

	upload the answers in Brightspace within the time limit. In case the regular on-campus exam can be held, no online possibility will be offered.
Exam Hours	3 hours

Year	**2022/2023**
Organization	**Aerospace Engineering**
Education	**Minors Aerospace Engineering**

LR-MI-232-22 Space Missions	
Minor Coordinator	Dr.ir. J. Bouwmeester

Year	**2022/2023**
Organization	**Aerospace Engineering**
Education	**Minors Aerospace Engineering**

Standard programme	
Minor Coordinator	Dr.ir. J. Bouwmeester

AE3522UL	Fundamentals of Astronomy	4
Responsible Instructor	M.A. Kenworthy	
Instructor	Dr. M. Rybak	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course provides 2 lectures/instructions per week of 2x45 minutes each. There are three subtopics consisting of Stars, Galaxies and Exoplanets. Each section lasts 2 weeks (with 4 lectures).	
Study Goals	The aim of this course is to give an overview of astronomical and astrophysical phenomena appropriate for the Space Missions Minor. The learning goals are: Understand the origin, evolution and end stages of stars Understand the structure and evolution of galaxies The evolution of the Universe and galaxies over cosmological timescales How we detect extrasolar planets and the search for life in the Universe	
Education Method	The content is provided in lectures throughout the course and with weekly homework sheets.	
Literature and Study Materials	Lecture notes available on BrightSpace. Astronomy papers that can be accessed online by the students.	
Prerequisites	This course is only available for students who are registered in the Space Missions minor.	
Assessment	Three Assignments (one on each of the topics of Galaxies, Stars and Exoplanets) will be given at the end of that subsection of lectures with a deadline one week after the end of the section. Each will count for 20% of the final grade, for a total of 60% of the final grade. An exam at the end of the course tests the knowledge and skills acquired during the course, which will count for 40% of the final grade.	

AE3523	Planetary Sciences	3
Responsible Instructor	Dr. S.J. de Vet	
Instructor	M.A. Kenworthy	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Planetary Science will offer you a perspective on the state-of-art in the field of planetary science performed inside the solar system. Past missions and insights from ongoing missions provide a unique perspective on the research performed in this field and new arising science questions. In this course we will reflect on the cross-disciplinary character of the science being performed at the interface of geoscience and astronomy.	
Study Goals	After completion of the course Planetary Science, you will be able to explain how we explore the Solar System and you will apply your scientific knowledge to describe characteristics of planets inside the Solar System and the exoplanets beyond. After this course the student is (cq. you are) able to: LO1: Explain how knowledge on planetary processes and evolution is obtained using planetary exploration missions. LO2: Perform calculations on planetary processes that have been observed inside the Solar System. LO3: Analyse key properties of planets and other planetary bodies inside the Solar System, using data from past and present satellite missions. LO4: Point out the underlying reasons for the physical differences between planets and other planetary bodies inside the Solar System.	
Education Method	Lectures (possibly pre-recorded with online/on-site option for Q&A after streaming of the lecture); two learning labs (i.e., homework assignments), with one dedicated to planetary processes and materials, and one focussed on icy moons, exomoons and habitability.	
Literature and Study Materials	The course will be based on parts of the book: McSween, Jr, H., Moersch, J., Burr, D., Dunne, W., Emery, J., Kah, L., & McCanta, M. (2019). Planetary Geoscience. Cambridge: Cambridge University Press. doi:10.1017/9781316535769 Lecture notes (slides) and assignments will be made available on BrightSpace.	
Assessment	The two learning labs are offered as a homework assignment via BrightSpace that will be graded. Feedback to the student will be provided after grading. A grade >6.0 is needed for each assignment to enter the final exam. For all students the final grade for the course will be based on a written exam (60%) and the grades for the digital learning labs (20% each). The exam will test all theory provided in the lectures, the literature (via self-study) and the skills practiced in the provided exercises.	
Enrolment / Application	This course is primarily intended for students who are registered in the Space Missions minor. If you would like to enrol for this course as elective outside this scope (e.g. for a 'free minor'), please send an email to course coordinator dr. Sebastiaan de Vet (s.j.devet@tudelft.nl) and motivate your request. The course coordinator will decide if you can be admitted.	

AE3524	Spacecraft Design and Astrodynamics	5

Responsible Instructor	Dr.ir. J. Bouwmeester
Instructor	Dr.ir. P.N.A.M. Visser
Instructor	Dr. J. Guo
Instructor	A. Cervone
Instructor	Dr. A. Menicucci
Instructor	S. Speretta
Instructor	I. Uriol Balbin
Contact Hours / Week x/x/x/x	6/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English

Course Contents

The course comprises 20 lectures. Some lectures have more emphasis on theory, some more on excercises. Below is a list of the lectures, the lecturers and its content. Please note that the order of the lectures is indicative and is subject to change. Up-to-date information will be provided on BrightSpace.

1. Spacecraft Basics - Jasper Bouwmeester
   Brief history of spaceflight
   Principles and examples of spacecraft
   Subsystem breakdown & functions

2. Astrodynamics 1/2 - Pieter Visser
   Brief history orbital mechanics
   Keplers law
   Newtons universal law of gravitation
   Restricted two-Body problem
   Orbital elements

3. Astrodynamics 2/2 - Pieter Visser
   Application orbits
   Ground tracks
   Orbit perturbations
   Orbit transfer

4. Structures 1/2 - Ines Uriol Balbin
   Spacecraft design loads
   Material selection for space missions
   Simulation of space structures
   Testing of space structures

5. Structures 2/2 - Ines Uriol Balbin
   Structural design
   Structural analysis

6. Command & Data Handling 1/2 - Alessandra Menicucci
   Intro to On-Board Data Handling (OBDH)
   Scope of an on-board computer
   Data handling system & unit architecture
   OBDH bus architecture
   OBDH operations modes

7. Command & Data Handling 2/2 - Alessandra Menicucci
   Designing an On-Board Computer
   Radiation effect mitigation
   Data protection
   On-Board Computer processor selection
   Memory selection

8. Electrical Power 1/2 - Jasper Bouwmeester
   Energy sources
   Solar power acquisition
   Voltage regulation
   Power distribution

9. Electrical Power 2/2 - Jasper Bouwmeester
   Preliminary sizing
   Design

10. Thermal Control - Jasper Bouwmeester
   Thermodynamic basics
   Passive control
   Active cooling & heating
   Sizing and design

11. Attitude Determination & Control 1/2 - Jasper Bouwmeester
   Orientation & reference frames
   Sensors overview & working principles
   Actuators overview & working principles
   Algorithms: general principles

12. Attitude Determination & Control 2/2 - Jasper Bouwmeester
   Preliminary sizing
   Design of sensors
   Design of actuators

	13. Propulsion 1/2 - Angelo Cervone Rocket performance equations Liquid and solid propellant engines 14. Propulsion 2/2 - Angelo Cervone Electric propulsion Advanced propulsion systems 15. Communication 1/2 - Stefano Speretta Basic link budget Main radio components Communication chain 16. Communication 2/2 - Stefano Speretta Link budget error and statistical analysis Introduction to radio-navigation 17. Guidance, Navigation & Control 1/2 - Jian Guo Guidance and trajectory planning Navigation Kinematics and dynamics Trajectory control 18. Guidance, Navigation & Control 2/2 - Jian Guo GNC system architecture design 19. Satellite Design 1/2 - Guest lecturer Example of state-of-the-art institutional satellite design 20. Satellite Design 2/2 Guest lecturer Example of state-of-the-art commercial satellite design
Study Goals	The aim of this course is to provide insight in the functions and design options of spacecraft and its astrodynamics. The learning goals are: Describe spacecraft orbits and calculate its main characteristics. Formulate the technical principles of spacecraft bus subsystems. Identify and compare the available subsystem options at component level, calculate their key characteristics and provide their physical and technical limitations. Characterize the key performance parameters of different spacecraft subsystems and compare the values obtained by ideal theory to the real-life ones. Perform preliminary design of a spacecraft subsystem and/or its astrodynamics based on key requirements and evaluate its performance.
Education Method	The content is provided in lectures (3 a week), of which some contain major exercises to performed, guided and explained in class. Students can self-study by reading annotated hand-outs and performing online exercises on BrightSpace. Students will perform an individual assignment in which they will perform preliminary design of a satellite subsystem or a satellite orbit. This assignment is provided and guided by the associated lectures.
Literature and Study Materials	Annotated lecture slides, available on BrightSpace. Book: Wertz and Larson, Space Mission Analysis & Design. In class 'large' exercises during instructions. Workouts made available after instructions. Small exercises on BrightSpace (MapleTA) for self-study.
Assessment	For all students, except AE students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. The exam will test all theory provided in the lectures and skills practiced in the provided exercises. In the assignment (28 hours total workload), the student will need to perform preliminary design of a subsystem or algorithm, including calculations/simulations, according to a set of provided technical requirements. Assessment will be based on a small report and a 30-minute discussion with the responsible lecturer on the results. For AE students: Given the partial overlap in the theory provided in the first two years of the aerospace bachelor, the examination will only count for 60% of the final grade. The assignment will be extended with additional objective/deliverable and will count for 40% of the final grade. Assessment will be based on a small report and a 45-minute discussion with the responsible lecturer on the results. For Astronomy students: Instead of the 4 ECTS astronomy course, which completely overlaps with content from the astronomy bachelor, students will perform two additional extended assignments (see AE students) next to a single regular assignment. The extended assignments are included in the 4 ECTS extended course (AE3524E), for which each assignment counts for 50% of its final grade. Students can provide their preferences for the assignments.
Enrolment / Application	This course is only available for students who are registered in the Space Missions minor. It is not possible to follow this course through other programs (free minor, Erasmus program, master, etcetera).

AE3526	Project Design and Analysis	2
Responsible Instructor	Dr. J.A.M. Vanhamel	
Instructor	Dr. A. Monreal Ibero	
Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Students follow courses on: Systems engineering fundamentals (2 lectures); Project management (1 lecture); Scientific analysis (1 lecture); Introduction to Collaborative Design Lab(CDL) (1 lecture). Additionally, each student team (consisting of about 8 students) will work individually on a Mission Assignment, provided by a Mission Client. The latter can be ESA, SRON, TNO, etc. This Mission Client will provide the mission objectives and top-level requirements. Student translate these into a top-level spacecraft and payload design, able to perform the scientific analysis.	
Study Goals	The students familiarize themselves with their given Mission Assignment, provided by an (external) Mission Client, interact with the Mission Client on the key requirements and perform a first scientific analysis and conceptual system design. The goal of the course is for the students to prepare a written report, describing the mission and its key scientific and engineering requirements, as well as the conceptual design. Additionally, the students also prepare an individual report, describing their personal contribution to the scientific case study and the engineering case study (objectives, methods and identifying the key interactions required).	
Education Method	Lectures will be given as indicated in the Course Contents. For the Mission Assignment; each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of the facility.	
Assessment	At the end of the course, each student team will present their work during a review session (presentation). The other teams, together with the Mission Client and members of the Academic Staff will assess the delivered work. More specific, after the review session, both Tutors (Scientific and Engineering), together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. A group report, describing the mission and its key scientific and engineering requirements, as well as the conceptual design will be written by the end of the course. In parallel with the group report, each student will write an individual report, reflecting the personal contribution to the presented work. Also this report will be written by the end of the course. A final score will be given by both Tutors (Scientific and Engineering), based on the presentation (30%), the group report (40%) and the individual report (30%).	
Enrolment / Application	For Space Missions minor students only	

AE3527UL	Scientific Case Study	4
Responsible Instructor	Dr. A. Monreal Ibero	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Using the preliminary scientific question and top-level requirements defined in the course AE3526 as input, the parameters of the required observations are determined and fine-tuned. Specifically, instrumental parameters such as the central observing wavelength, the total bandwidth of the observations, the spectral resolution, spatial sampling, field of view, or required temporal sampling are considered. These parameters then determine the choice of telescope aperture (to gather enough photons) and detector technology used. In short, the students define the optical configuration required to meet the science goals.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary scientific case and key question to be answered, as well as top level requirements for the instrumentation in the spacecraft or space system. The goal of the course is to make the students more familiar with the processes involved in the definition of the scientific aspects of a spatial mission. In particular, the students will find and review the relevant literature on a given science topic. They will need to get a general picture on the science topic and a more detailed insight of the specific question(s) they will address with the mission. Likewise, they will identify similar state-of-the-art missions (or ground-based facilities), and determine and highlight the unique features of this mission. Then, they will establish and investigate the physical relationships between the science target, the optomechanics within the spacecraft, and the resultant detection and processing of the detected signal. In the process, the low level scientific and technical requirements from the initial mission statement and requirements, are defined and refined in an iterative manner.	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of such a facility. At the Leiden University premises, the students will work at The Studio Classroom in the Hugyens building.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff participate in these reviews. More specifically, the global assessment will be the combination of the outcome of the following 4 items: 1. An initial presentation review session. After that session, the Scientific Tutor, together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. The results will be taken into account for the final grading for an amount of 10%. 2. A group report written at the end of the course, describing the scientific context, immediate science goals, need for a mission, top level requirements for the payload, and low-level detailed solution. This will constitute 40% of the final individual grade. 3. A document submitted by each individual student at the same time as the group report, where she/he will present individually a reflection on the personal contribution to the presented scientific case study. This will count up to 30% of the final grade. 4. A final presentation, where the team will present their project to the other teams, the Mission Clients, tutors, and Responsible of the course. It will be graded in the same manner as the initial presentation and will contribute up to a 20% of the total grade.	
Enrolment / Application	For Space Missions minor students only	

AE3528	Engineering Case Study	4
Responsible Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Using well selected systems engineering tools, engineering analysis and design tools and online research, the spacecraft or space system will be designed in more detail. The preliminary design obtained in the course AE3526 will be used as input. Technical requirements will be updated and derived with more detail for each subsystem. Key options for subsystems and components will be identified, either commercially available or custom designed. A trade-off for each will be performed and justified. All system budgets will close and interfaces are compatible, as proven in tables and figures. The report will follow a standard template provided for the course.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary design of the spacecraft or space system. The goal of the course is to make the students more familiar with the design process of a spacecraft. More specific, the spacecraft requirements to be set shall be driven by the functional analysis of the spacecraft and the mission profile. Based on this, the initial spacecraft design can be done (sizing, budgeting, etc.). As a last step, the different spacecraft subsystems are designed (analysis and architecture).	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of such a facility.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff will participate in these reviews. More specific, after the review session, the Engineering Tutor, together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. The results will be taken into account for the final grading for an amount of 30%. By the end of the course, a group report describing the selected engineering design and the matching of the selected design in the frame of the Mission Assignment, is delivered. The quality of the whole report, based on the trade-offs, technical analysis using adequate systems engineering tools, design description as well as the introduction, conclusions and overall quality, contributes 40% to the final grade. Additionally, in parallel to the group report, an individual report written by each student, has to be delivered by the end of the course. This report describes the individual contribution to the Engineering Case Study from a technical point-of-view, as well as a self-reflection on the learned process. This deliverable counts for 30% of the final grade. A final score will be given, based on the review sessions (30%), the group report (40%) and the individual report (30%).	
Enrolment / Application	For Space Missions minor students only	

AE3529	Space Instrumentation	5
Responsible Instructor	Dr. J.J.D. Loicq	
Instructor	Dr. J.A.M. Vanhamel	
Instructor	M.A. Kenworthy	
Instructor	J.G. De Teixeira da Encarnacao	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Space instruments are the links between the science target, the missions, the data-users, the spacecraft, The goal of this course will be to give students insights into the capabilities of instrumentation and how they contribute to the observational strategies of the mission. Space instrumentation conception is based on very multidisciplinary technics and technologies. Optics routes photons to detectors, when mechanics gives the stability and holds optical surfaces, detectors, radiators, electronics, while thermic is given by environment (orbits, spacecraft,) and apply constraints. The conception of the instruments in a system engineering approach will be discussed. Space Instrumentation can be subdivided into two main groups: remote and in situ instrumentations. Both will be covered during this course. The space instrumentation course will also give an overview of the already existing and under development instruments. The relationship between instruments and their associated space mission will be taught.	
Study Goals	The students should understand and describes The links between instruments, mission and observation strategies The physical observables (light, ions, magnetic/electric fields and related phenomena) the space instruments are measuring. The different type of space instruments. How they are designed and built The effect of the space environment on the instrument designs The keys drivers of a space instruments	
Education Method	The content is provided in lectures, which include some exercises that will be performed, guided , and explained. Students can self-study by reading lecture notes and performing online exercises on BrightSpace. Students will perform an individual assignment in in which they analyse an existing space mission.	
Assessment	For all students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. Exam (80%): The exam will test all theories provided in the lectures and skills practiced in the provided exercises. Assignment (20%): In the assignment, it will be asked to the students to provide a critical study of a space mission by the mean an analysis of (at least) one scientific article related to a mission (e.g. Space Science Review, JATIS,) The objective will be to understand the space instrument chosen and its relationship with the mission, and their observation strategies. Assessment will be based on a small report (max 6 pages).	

CT3533	Earth Observation	3
Responsible Instructor	Prof.dr.ir. R.F. Hanssen	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Earth observation is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Remote sensing observations from spaceborne instrument inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. This course aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain bio-& geophysical variables iii) to evaluate the different advantages/disadvantages of earth observation technologies. This should allow students to understand how to formulate user requirements for understanding/monitoring Earth into an earth observation mission requirements.	
Study Goals	The overall goal of this course is to provide insight into the physical principles of Earth observation and how earth observation technologies can be used to assess and monitor geo- & biophysical variables on Earth. After this course the student is (cq. you are) able to: LO1: Summarise the signals of the earth in the different parts of the electromagnetic (EM) spectrum and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and subsurface). LO2: Characterise the different observable signals and parameters from Earth and how they can be observed using different remote sensing technologies. LO3: Assess remote sensing technologys advantages/disadvantages for observing geo- & biophysical variables on Earth. LO4: Translate user requirements into mission requirements for earth observation mission. LO5: Evaluate a basic remote sensing analysis to retrieve a geo- & biophysical variable from Earth Observation data.	
Education Method	Classical lectures complemented with pre-recorded videos to maximize interaction in class (flip-the-classroom). Two lab sessions to work on assignment in groups with peer-review.	
Assessment	Written open book exam (70%) Assignment (30%) in groups of 4 where students need to translate user requirements into mission requirements for a specific earth observation application. Students need to score minimal a 6 on each component to pass the course.	
Expected prior Knowledge	No specific knowledge required.	
Academic Skills	-	
Literature & Study Materials	Video recordings Book: Rees (2012). Physical Principles of Remote Sensing https://doi.org/10.1017/CBO9781139017411	
Judgement	Students need to score minimal a 6 on each component (exam/assignment) to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Open book exam where lecture notes, lecture slides, book and calculator are permitted.	
Collegerama	No	

Year	**2022/2023**
Organization	**Aerospace Engineering**
Education	**Minors Aerospace Engineering**

Programme for Astronomy students (Leiden University)	
Minor Coordinator	Dr.ir. J. Bouwmeester

AE3523	Planetary Sciences	3
Responsible Instructor	Dr. S.J. de Vet	
Instructor	M.A. Kenworthy	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course Planetary Science will offer you a perspective on the state-of-art in the field of planetary science performed inside the solar system. Past missions and insights from ongoing missions provide a unique perspective on the research performed in this field and new arising science questions. In this course we will reflect on the cross-disciplinary character of the science being performed at the interface of geoscience and astronomy.	
Study Goals	After completion of the course Planetary Science, you will be able to explain how we explore the Solar System and you will apply your scientific knowledge to describe characteristics of planets inside the Solar System and the exoplanets beyond. After this course the student is (cq. you are) able to: LO1: Explain how knowledge on planetary processes and evolution is obtained using planetary exploration missions. LO2: Perform calculations on planetary processes that have been observed inside the Solar System. LO3: Analyse key properties of planets and other planetary bodies inside the Solar System, using data from past and present satellite missions. LO4: Point out the underlying reasons for the physical differences between planets and other planetary bodies inside the Solar System.	
Education Method	Lectures (possibly pre-recorded with online/on-site option for Q&A after streaming of the lecture); two learning labs (i.e., homework assignments), with one dedicated to planetary processes and materials, and one focussed on icy moons, exomoons and habitability.	
Literature and Study Materials	The course will be based on parts of the book: McSween, Jr, H., Moersch, J., Burr, D., Dunne, W., Emery, J., Kah, L., & McCanta, M. (2019). Planetary Geoscience. Cambridge: Cambridge University Press. doi:10.1017/9781316535769 Lecture notes (slides) and assignments will be made available on BrightSpace.	
Assessment	The two learning labs are offered as a homework assignment via BrightSpace that will be graded. Feedback to the student will be provided after grading. A grade >6.0 is needed for each assignment to enter the final exam. For all students the final grade for the course will be based on a written exam (60%) and the grades for the digital learning labs (20% each). The exam will test all theory provided in the lectures, the literature (via self-study) and the skills practiced in the provided exercises.	
Enrolment / Application	This course is primarily intended for students who are registered in the Space Missions minor. If you would like to enrol for this course as elective outside this scope (e.g. for a 'free minor'), please send an email to course coordinator dr. Sebastiaan de Vet (s.j.devet@tudelft.nl) and motivate your request. The course coordinator will decide if you can be admitted.	

AE3524	Spacecraft Design and Astrodynamics	5
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	

Course Contents

The course comprises 20 lectures. Some lectures have more emphasis on theory, some more on excercises. Below is a list of the lectures, the lecturers and its content. Please note that the order of the lectures is indicative and is subject to change. Up-to-date information will be provided on BrightSpace.

1. Spacecraft Basics - Jasper Bouwmeester
   Brief history of spaceflight
   Principles and examples of spacecraft
   Subsystem breakdown & functions

2. Astrodynamics 1/2 - Pieter Visser
   Brief history orbital mechanics
   Keplers law
   Newtons universal law of gravitation
   Restricted two-Body problem
   Orbital elements

3. Astrodynamics 2/2 - Pieter Visser
   Application orbits
   Ground tracks
   Orbit perturbations
   Orbit transfer

4. Structures 1/2 - Ines Uriol Balbin
   Spacecraft design loads
   Material selection for space missions
   Simulation of space structures
   Testing of space structures

5. Structures 2/2 - Ines Uriol Balbin
   Structural design
   Structural analysis

6. Command & Data Handling 1/2 - Alessandra Menicucci
   Intro to On-Board Data Handling (OBDH)
   Scope of an on-board computer
   Data handling system & unit architecture
   OBDH bus architecture
   OBDH operations modes

7. Command & Data Handling 2/2 - Alessandra Menicucci
   Designing an On-Board Computer
   Radiation effect mitigation
   Data protection
   On-Board Computer processor selection
   Memory selection

8. Electrical Power 1/2 - Jasper Bouwmeester
   Energy sources
   Solar power acquisition
   Voltage regulation
   Power distribution

9. Electrical Power 2/2 - Jasper Bouwmeester
   Preliminary sizing
   Design

10. Thermal Control - Jasper Bouwmeester
   Thermodynamic basics
   Passive control
   Active cooling & heating
   Sizing and design

11. Attitude Determination & Control 1/2 - Jasper Bouwmeester
   Orientation & reference frames
   Sensors overview & working principles
   Actuators overview & working principles
   Algorithms: general principles

12. Attitude Determination & Control 2/2 - Jasper Bouwmeester
   Preliminary sizing
   Design of sensors
   Design of actuators

	13. Propulsion 1/2 - Angelo Cervone Rocket performance equations Liquid and solid propellant engines 14. Propulsion 2/2 - Angelo Cervone Electric propulsion Advanced propulsion systems 15. Communication 1/2 - Stefano Speretta Basic link budget Main radio components Communication chain 16. Communication 2/2 - Stefano Speretta Link budget error and statistical analysis Introduction to radio-navigation 17. Guidance, Navigation & Control 1/2 - Jian Guo Guidance and trajectory planning Navigation Kinematics and dynamics Trajectory control 18. Guidance, Navigation & Control 2/2 - Jian Guo GNC system architecture design 19. Satellite Design 1/2 - Guest lecturer Example of state-of-the-art institutional satellite design 20. Satellite Design 2/2 Guest lecturer Example of state-of-the-art commercial satellite design
Study Goals	The aim of this course is to provide insight in the functions and design options of spacecraft and its astrodynamics. The learning goals are: Describe spacecraft orbits and calculate its main characteristics. Formulate the technical principles of spacecraft bus subsystems. Identify and compare the available subsystem options at component level, calculate their key characteristics and provide their physical and technical limitations. Characterize the key performance parameters of different spacecraft subsystems and compare the values obtained by ideal theory to the real-life ones. Perform preliminary design of a spacecraft subsystem and/or its astrodynamics based on key requirements and evaluate its performance.
Education Method	The content is provided in lectures (3 a week), of which some contain major exercises to performed, guided and explained in class. Students can self-study by reading annotated hand-outs and performing online exercises on BrightSpace. Students will perform an individual assignment in which they will perform preliminary design of a satellite subsystem or a satellite orbit. This assignment is provided and guided by the associated lectures.
Literature and Study Materials	Annotated lecture slides, available on BrightSpace. Book: Wertz and Larson, Space Mission Analysis & Design. In class 'large' exercises during instructions. Workouts made available after instructions. Small exercises on BrightSpace (MapleTA) for self-study.
Assessment	For all students, except AE students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. The exam will test all theory provided in the lectures and skills practiced in the provided exercises. In the assignment (28 hours total workload), the student will need to perform preliminary design of a subsystem or algorithm, including calculations/simulations, according to a set of provided technical requirements. Assessment will be based on a small report and a 30-minute discussion with the responsible lecturer on the results. For AE students: Given the partial overlap in the theory provided in the first two years of the aerospace bachelor, the examination will only count for 60% of the final grade. The assignment will be extended with additional objective/deliverable and will count for 40% of the final grade. Assessment will be based on a small report and a 45-minute discussion with the responsible lecturer on the results. For Astronomy students: Instead of the 4 ECTS astronomy course, which completely overlaps with content from the astronomy bachelor, students will perform two additional extended assignments (see AE students) next to a single regular assignment. The extended assignments are included in the 4 ECTS extended course (AE3524E), for which each assignment counts for 50% of its final grade. Students can provide their preferences for the assignments.
Enrolment / Application	This course is only available for students who are registered in the Space Missions minor. It is not possible to follow this course through other programs (free minor, Erasmus program, master, etcetera).

AE3524E	Spacecraft Design and Astrodynamics Extended	4
Responsible Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE3524	
Course Contents	This is the extension module for AE3524 Spacecraft Design & Astrodynamics.	
Study Goals	Perform preliminary design of a spacecraft subsystem and/or its astrodynamics based on key requirements and evaluate its performance	
Education Method	Only excercises. See AE3524 for more information.	
Assessment	Two extended assignments with a studyoad of 56 hours each which count for 50% of the final grade. Students can provide their preferences for the assignments.	
Enrolment / Application	This course is only available for students who are registered in the Space Missions minor and follow an Astronomy bachelor at the University of Leiden. It is not available for other students.	

AE3526	Project Design and Analysis	2
Responsible Instructor	Dr. J.A.M. Vanhamel	
Instructor	Dr. A. Monreal Ibero	
Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Students follow courses on: Systems engineering fundamentals (2 lectures); Project management (1 lecture); Scientific analysis (1 lecture); Introduction to Collaborative Design Lab(CDL) (1 lecture). Additionally, each student team (consisting of about 8 students) will work individually on a Mission Assignment, provided by a Mission Client. The latter can be ESA, SRON, TNO, etc. This Mission Client will provide the mission objectives and top-level requirements. Student translate these into a top-level spacecraft and payload design, able to perform the scientific analysis.	
Study Goals	The students familiarize themselves with their given Mission Assignment, provided by an (external) Mission Client, interact with the Mission Client on the key requirements and perform a first scientific analysis and conceptual system design. The goal of the course is for the students to prepare a written report, describing the mission and its key scientific and engineering requirements, as well as the conceptual design. Additionally, the students also prepare an individual report, describing their personal contribution to the scientific case study and the engineering case study (objectives, methods and identifying the key interactions required).	
Education Method	Lectures will be given as indicated in the Course Contents. For the Mission Assignment; each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of the facility.	
Assessment	At the end of the course, each student team will present their work during a review session (presentation). The other teams, together with the Mission Client and members of the Academic Staff will assess the delivered work. More specific, after the review session, both Tutors (Scientific and Engineering), together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. A group report, describing the mission and its key scientific and engineering requirements, as well as the conceptual design will be written by the end of the course. In parallel with the group report, each student will write an individual report, reflecting the personal contribution to the presented work. Also this report will be written by the end of the course. A final score will be given by both Tutors (Scientific and Engineering), based on the presentation (30%), the group report (40%) and the individual report (30%).	
Enrolment / Application	For Space Missions minor students only	

AE3527UL	Scientific Case Study	4
Responsible Instructor	Dr. A. Monreal Ibero	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Using the preliminary scientific question and top-level requirements defined in the course AE3526 as input, the parameters of the required observations are determined and fine-tuned. Specifically, instrumental parameters such as the central observing wavelength, the total bandwidth of the observations, the spectral resolution, spatial sampling, field of view, or required temporal sampling are considered. These parameters then determine the choice of telescope aperture (to gather enough photons) and detector technology used. In short, the students define the optical configuration required to meet the science goals.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary scientific case and key question to be answered, as well as top level requirements for the instrumentation in the spacecraft or space system. The goal of the course is to make the students more familiar with the processes involved in the definition of the scientific aspects of a spatial mission. In particular, the students will find and review the relevant literature on a given science topic. They will need to get a general picture on the science topic and a more detailed insight of the specific question(s) they will address with the mission. Likewise, they will identify similar state-of-the-art missions (or ground-based facilities), and determine and highlight the unique features of this mission. Then, they will establish and investigate the physical relationships between the science target, the optomechanics within the spacecraft, and the resultant detection and processing of the detected signal. In the process, the low level scientific and technical requirements from the initial mission statement and requirements, are defined and refined in an iterative manner.	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of such a facility. At the Leiden University premises, the students will work at The Studio Classroom in the Hugyens building.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff participate in these reviews. More specifically, the global assessment will be the combination of the outcome of the following 4 items: 1. An initial presentation review session. After that session, the Scientific Tutor, together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. The results will be taken into account for the final grading for an amount of 10%. 2. A group report written at the end of the course, describing the scientific context, immediate science goals, need for a mission, top level requirements for the payload, and low-level detailed solution. This will constitute 40% of the final individual grade. 3. A document submitted by each individual student at the same time as the group report, where she/he will present individually a reflection on the personal contribution to the presented scientific case study. This will count up to 30% of the final grade. 4. A final presentation, where the team will present their project to the other teams, the Mission Clients, tutors, and Responsible of the course. It will be graded in the same manner as the initial presentation and will contribute up to a 20% of the total grade.	
Enrolment / Application	For Space Missions minor students only	

AE3528	Engineering Case Study	4
Responsible Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Using well selected systems engineering tools, engineering analysis and design tools and online research, the spacecraft or space system will be designed in more detail. The preliminary design obtained in the course AE3526 will be used as input. Technical requirements will be updated and derived with more detail for each subsystem. Key options for subsystems and components will be identified, either commercially available or custom designed. A trade-off for each will be performed and justified. All system budgets will close and interfaces are compatible, as proven in tables and figures. The report will follow a standard template provided for the course.	
Study Goals	During the Project Design and Analysis course (see AE3526), each student team worked out an approach on the first scientific analysis and conceptual system design. Based on this, each team provides a preliminary design of the spacecraft or space system. The goal of the course is to make the students more familiar with the design process of a spacecraft. More specific, the spacecraft requirements to be set shall be driven by the functional analysis of the spacecraft and the mission profile. Based on this, the initial spacecraft design can be done (sizing, budgeting, etc.). As a last step, the different spacecraft subsystems are designed (analysis and architecture).	
Education Method	Each student team will work individually on their assignment. Tutors (scientific and engineering) will guide the students throughout the process. Students will partially use the Collaborative Design Lab (CDL) in order to experience the use of such a facility.	
Assessment	Throughout the course, several review processes will take place. The other teams, the Mission Client and members of the Academic Staff will participate in these reviews. More specific, after the review session, the Engineering Tutor, together with the Responsible, will grade the students based on the inputs of the Mission Client, the in-depth analysis of the assignment, the outcome and the overall quality of the presentation. The results will be taken into account for the final grading for an amount of 30%. By the end of the course, a group report describing the selected engineering design and the matching of the selected design in the frame of the Mission Assignment, is delivered. The quality of the whole report, based on the trade-offs, technical analysis using adequate systems engineering tools, design description as well as the introduction, conclusions and overall quality, contributes 40% to the final grade. Additionally, in parallel to the group report, an individual report written by each student, has to be delivered by the end of the course. This report describes the individual contribution to the Engineering Case Study from a technical point-of-view, as well as a self-reflection on the learned process. This deliverable counts for 30% of the final grade. A final score will be given, based on the review sessions (30%), the group report (40%) and the individual report (30%).	
Enrolment / Application	For Space Missions minor students only	

AE3529	Space Instrumentation	5
Responsible Instructor	Dr. J.J.D. Loicq	
Instructor	Dr. J.A.M. Vanhamel	
Instructor	M.A. Kenworthy	
Instructor	J.G. De Teixeira da Encarnacao	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Space instruments are the links between the science target, the missions, the data-users, the spacecraft, The goal of this course will be to give students insights into the capabilities of instrumentation and how they contribute to the observational strategies of the mission. Space instrumentation conception is based on very multidisciplinary technics and technologies. Optics routes photons to detectors, when mechanics gives the stability and holds optical surfaces, detectors, radiators, electronics, while thermic is given by environment (orbits, spacecraft,) and apply constraints. The conception of the instruments in a system engineering approach will be discussed. Space Instrumentation can be subdivided into two main groups: remote and in situ instrumentations. Both will be covered during this course. The space instrumentation course will also give an overview of the already existing and under development instruments. The relationship between instruments and their associated space mission will be taught.	
Study Goals	The students should understand and describes The links between instruments, mission and observation strategies The physical observables (light, ions, magnetic/electric fields and related phenomena) the space instruments are measuring. The different type of space instruments. How they are designed and built The effect of the space environment on the instrument designs The keys drivers of a space instruments	
Education Method	The content is provided in lectures, which include some exercises that will be performed, guided , and explained. Students can self-study by reading lecture notes and performing online exercises on BrightSpace. Students will perform an individual assignment in in which they analyse an existing space mission.	
Assessment	For all students, the final grade will be for 80% based on a written exam and 20% based on an individual design assignment. Exam (80%): The exam will test all theories provided in the lectures and skills practiced in the provided exercises. Assignment (20%): In the assignment, it will be asked to the students to provide a critical study of a space mission by the mean an analysis of (at least) one scientific article related to a mission (e.g. Space Science Review, JATIS,) The objective will be to understand the space instrument chosen and its relationship with the mission, and their observation strategies. Assessment will be based on a small report (max 6 pages).	

CT3533	Earth Observation	3
Responsible Instructor	Prof.dr.ir. R.F. Hanssen	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Earth observation is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Remote sensing observations from spaceborne instrument inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. This course aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain bio-& geophysical variables iii) to evaluate the different advantages/disadvantages of earth observation technologies. This should allow students to understand how to formulate user requirements for understanding/monitoring Earth into an earth observation mission requirements.	
Study Goals	The overall goal of this course is to provide insight into the physical principles of Earth observation and how earth observation technologies can be used to assess and monitor geo- & biophysical variables on Earth. After this course the student is (cq. you are) able to: LO1: Summarise the signals of the earth in the different parts of the electromagnetic (EM) spectrum and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and subsurface). LO2: Characterise the different observable signals and parameters from Earth and how they can be observed using different remote sensing technologies. LO3: Assess remote sensing technologys advantages/disadvantages for observing geo- & biophysical variables on Earth. LO4: Translate user requirements into mission requirements for earth observation mission. LO5: Evaluate a basic remote sensing analysis to retrieve a geo- & biophysical variable from Earth Observation data.	
Education Method	Classical lectures complemented with pre-recorded videos to maximize interaction in class (flip-the-classroom). Two lab sessions to work on assignment in groups with peer-review.	
Assessment	Written open book exam (70%) Assignment (30%) in groups of 4 where students need to translate user requirements into mission requirements for a specific earth observation application. Students need to score minimal a 6 on each component to pass the course.	
Expected prior Knowledge	No specific knowledge required.	
Academic Skills	-	
Literature & Study Materials	Video recordings Book: Rees (2012). Physical Principles of Remote Sensing https://doi.org/10.1017/CBO9781139017411	
Judgement	Students need to score minimal a 6 on each component (exam/assignment) to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Open book exam where lecture notes, lecture slides, book and calculator are permitted.	
Collegerama	No	

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